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Full attitude state reconstruction of tumbling space debris TOPEX/Poseidon via light-curve inversion with Quanta Photogrammetry

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ABSTRACT

The tumbling motion of defunct satellite TOPEX/Poseidon (T/P) is governed by the μ-scale Solar Radiation Pressure (SRP) torque that spun up the satellite from the initial, nadir-pointing to the current fast spinning state [1]. In this paper, we advance the methods of Quanta Photogrammetry (QPM) in an effort to determine satellite attitude parameters – the inertial orientation of the body angular momentum vector $\boldsymbol{L}$, the rotational phase and rate as well as the object pole coordinates with respect to $\boldsymbol{L}$; the determined pole offset is 3.52°. The presented data support the hypothesis that the full characteristics of the satellite attitude dynamics can be obtained from a single radiant photogram measured by the photon-counting system of the Graz Satellite Laser Ranging (SLR) Observatory during an overhead pass.

The analysis of the observed radiant signature of T/P gives the elevation angle of the normal vector to the solar array front surface at -1.7°, RMS $= 0.63^\circ$, in the spacecraft body-centered reference frame. This information, along with the pole position, is used to model the SRP forces and torques on the satellite and determine the T/P moment of inertia $I = 69728$, $\sigma_I = 75.2$ $[kgm^2]$, by best-fitting long-term spin simulation to the observed trend.

1. Introduction

1.1. Tumbling dynamics of defunct TOPEX/Poseidon

Ocean TOPography EXperiment TOPEX/Poseidon (T/P, COSPAR ID 1992-052 A, NORAD 22076) was a joint satellite altimeter mission between NASA and CNES designed to explore ocean circulation and its interaction with the atmosphere [2]. The satellite successfully operated for over 13 years on a near-circular orbit of 1340 km altitude and inclination 66°. The onboard altimeters measured the distance to the instantaneous sea surface with a precision of about 2 cm and helped to better understand global climate change and weather phenomena such as El Niño. Three independent techniques (SLR, DORIS, and GPS) were used to determine the satellite altitude for the altimeter calibration.

T/P has been decommissioned since January 5, 2006 becoming a passive space debris object with the dynamics governed by its interaction with the environmental forces and torques caused mainly by the Earth's gravitational and magnetic fields, as well as solar irradiation and both Earth albedo and thermal re-radiation. The long time series of the Satellite Laser Ranging (SLR) observations indicate that the T/P spin axis orientation remains in the vicinity of the orbital eccentricity vector (the Laplace-Runge-Lenz vector) with the detectable precession and nutation periodicities of 173.3 days and 101.6 days respectively [1]. Such complex tumbling dynamics, synchronized with the precession of the orbital plane, is related to the gravitational forces and torques acting on the dynamical oblateness (the difference in the principal moments of inertia) of the spacecraft; this phenomenon has been previously observed with the geodetic satellite Ajisai [3].

The combined photometric and SLR data collected over an 11-year time span indicate an increasing trend in the T/P spin energy and a relatively short rotational period of 10.73 s in October 2016. The direct solar radiation pressure has been identified as the main environmental factor responsible for the steady spin-up of the body. The research results presented by Kudak et al. [4] agree with, and complement, the SLR-based spin analysis and support the understanding of the observed attitude behavior.

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1.2. Optical methods for attitude observation of passive satellites

The optical tracking technologies, Satellite Laser Ranging (SLR) and photometry, do not require any form of cooperation with the space objects and thus are recommended for observation of the fully passive satellites. The SLR determines an absolute distance between a ground system reference point and the satellites by measuring the time-of-flight of ultra-short laser pulses transmitted towards the target and retro-reflected back to the ground detection system. The range measurements to geodetic satellites are used for precise orbit determination, the study of tectonic plate motion, Earth orientation and rotation parameters, as well as the determination of the gravity field and the geocenter position [5]. The low-power, high-repetition rate SLR efficiently performs day- and night-time measurements of the attitude parameters of cooperative satellites equipped with the retroreflective components such as the corner cube reflectors (CCR) or the Luneburg lenses [6–9]. The high-power Debris Laser Ranging (DLR) has been successfully used to determine orientation of non-cooperative objects such as tumbling rocket bodies or larger box-wing type payloads [10–13]. The Graz SLR station has recently demonstrated the innovative capability of high-power laser ranging to the non-cooperative satellites during daylight [14].

The photometric observation of sunlit space objects can be conducted at night-time and delivers time series of satellite brightness known as light curves [15]. Typically, the photometric samples are processed as a one-dimensional time series, where the temporal variation of an optical flux intensity can be analyzed by two distinctive categories of the epoch and amplitude methods. The epoch analysis estimates the periodicities of the photometric patterns and is represented by the Fast Fourier Transform or the Lomb-Scargle Periodogram [16] which estimates a frequency spectrum of unequally spaced data based on the least squares fit of sinusoids to the data samples. The Phase Dispersion Minimization [17] allows for the determination of relatively long rotational periods (in relation to the pass duration) by minimizing variance of the phase folded time series. The amplitude methods, successfully exercised by Williams [18] or Yanagisawa and Kurosaki [19] analyze ratio of the flux intensity extrema for deriving satellite attitude parameters.

Quanta Photogrammetry (QPM) has been introduced by Kucharski et al. [20] as a method of projecting single-photon detection events onto the phase bisector vector expressed in the satellite body-fixed reference frame [21]. As such, the method removes the time dependency from a light curve and creates a static photometric pattern over the angular coordinates (azimuth and elevation) of the satellite-centered and -fixed reference frame. The initial publication reported on the optical mapping of the small-scale surface structural anomalies of the fast-spinning satellite Ajisai, while the current paper demonstrates application of QPM to the complex problem of the space debris attitude determination through the radiant Signal-to-Noise ratio analysis.

Accurate attitude models can improve the long-term conjunction analyses [22] and mitigate risk of the close approach situations such as the one that occurred between T/P and Jason-2 on June 20, 2017 as reported by the ILRS Space Debris Study Group [23]. The attitude parameters are important in planning and modeling the Rendezvous and Proximity Operation for the potential Active Debris Removal mission [24], as well as in the development and optimization of the laser force de-spin and de-orbit maneuver scenarios [25].

2. Quanta Photogrammetry of TOPEX/Poseidon

2.1. Hypertemporal photometry at single-photon resolution

The Graz SLR station operates multiple optical techniques for satellite observations – a high repetition rate 2 kHz, 532 nm laser system allows for mm-accuracy range measurements to cooperative satellites and space debris, while a high-power laser is used for ranging to non-cooperative space objects, even during daylight [10,14]. The ultra-high 1 MHz repetition rate laser ranging up to the Geosynchronous Earth Orbit has been also successfully demonstrated [26]. The photon counting system, implemented in 2015 [27], measures light curves of sunlit satellites at the hypertemporal sampling rates over the spectral bandwidth of 780–900 nm [28]. The incoming solar photons collected by the receiver telescope are detected by the Single-Photon Avalanche Diode (τ-SPAD FAST) and converted into an electronic signal that is processed and time stamped in UTC by the Field Programmable Gate Array (FPGA). The acquired data is then binned and converted into a light curve for further analysis.

In orbit, the particles of direct solar irradiation can impact the satellite surface elements and interact with the materials in multiple ways including reflection, absorption and transmission. The interaction between an incoming photon and the surface element is characterized by a momentum transfer known as the solar radiation pressure effect [29]; small in magnitude, but acting continuously, it dominates the tumbling dynamics of T/P [1].

A free space optical link between the satellite and the receiver telescope effectively works as an information transmitter of the reflected solar photons through the turbulent atmospheric layers where the optical flux attenuates due to absorption by ozone, water vapor, and carbon dioxide as well as being scattered by air molecules, water and dust particles. The number of detected photons per time interval is expressed as a photon count rate in particles-per-second (Hz) and represents the intensity of the incoming light. An equivalent definition of this property is radiant flux which can be estimated by taking into account the wavelength dependent energy of the arriving photons per unit time and the spectral quantum efficiency of the τ-SPAD detector. This problem has been previously discussed as a part of the solar link budget analysis of specularly reflective satellite Ajisai [30].

Fig. 1 presents a sample of a T/P hypertemporal light curve observed by the single-photon detection system at Graz on June 7, 2015. The photometric timeseries presents modulation of the received optical signal due to satellite rotation about its center-of-mass at the period of 11.4 s. The pattern consists of a mixture of specular and diffuse reflections off of the tumbling satellite, with the specular features characterized by the sharp and relatively intense spikes in the flux intensity. The photon counting detectors do not integrate the incoming signal allowing for extremely high tempo-spatial sampling resolution of the specular flashes produced by various materials of space structures [31]. The limitation of the single-pixel SPAD detector is, however, that it covers entire field-of-view (FOV) of the receiver optics and thus it cannot separate the point-source signal photons from the noise of the background sky. A solution to this problem will come with the new generation Quanta Image Sensor (QIS) that offers high-efficiency photon counting capabilities at the megapixel resolution [32].

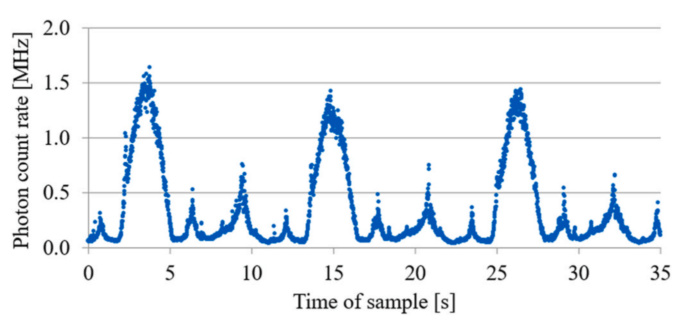

Fig. 1. A sample of the hypertemporal light curve of T/P collected by Graz on June 7, 2015. The photometric pattern represents 3 rotations of the satellite about its center-of-mass at the spin period of 11.4 s.

2.2. Solar optical link

Transmission of a photon from the radiation source (Sun) towards the detector (a terrestrial observer) through the single reflection point on satellite surface occurs at a specific geometrical configuration between the transmitter, reflector and receiver. At the instant of reflection, a geometrical system can be defined that determines properties of the satellite-observer optical link. The key components of this system are the satellite-centered unit direction vectors towards the Sun $\boldsymbol{S}$ and the receiver telescope $\boldsymbol{T}$, as well as their product - the phase vector $\boldsymbol{P} = \frac{\boldsymbol{S}+\boldsymbol{T}}{|\boldsymbol{S}+\boldsymbol{T}|}$. As illustrated in Fig. 2, the phase vector $\boldsymbol{P}$ bisects the phase angle p between the two direction vectors and creates an inclination angle $i = \arccos(\boldsymbol{P}\cdot\boldsymbol{N})$ with the surface normal vector $\boldsymbol{N}$. In the case of a flat surface element, the inclination angle must be lower than the angular radius of the solar disk ($\sim 0.26^\circ$) for the specular flash to occur in the observer direction.

An optical link between the reflective element and the ground observer can form when the aspect angles β, γ between the surface normal vector $\boldsymbol{N}$ and the corresponding direction vectors $\boldsymbol{S}$ and $\boldsymbol{T}$ remain lower than 90°. The incident photons can be absorbed (α), specularly reflected (ρ) or diffused (δ) by the illuminated surface at the fractional coefficients related through the equation $\alpha+\rho+\delta = 1$. The intensity of reflection depends on the on-axis optical reflector gain which for Lambertian diffusion can be modelled as:

$$G_\delta = \frac{\Omega_R}{\pi}\cos\beta\cos\gamma \tag{1}$$

where Ω_R is the solid angle of the reflective surface area. The specular reflection gain can be defined as a ratio of the surface element Ω_R and the ground reflection footprint Ω_F solid angles:

$$G_\rho = \frac{\Omega_R}{\Omega_F} \tag{2}$$

An observed intensity of solar reflection is a function of multiple variables related to the satellite optical properties, elevation-dependent atmospheric attenuation effects, and the parameters of the ground detection system. A detected photon flux Φ_q can be expressed as the photoelectron counts per second and modelled by:

$$\Phi_q = G_R\eta_R\sum_{\lambda=\lambda_0}^{\lambda_1}\eta_q(\lambda)\Phi_S(\lambda)A_D\eta_D\eta_A(\lambda) \tag{3}$$

where the photodetector response is integrated over the spectral bandwidth from λ_0 to λ_1 with the wavelength dependent quantum efficiency $\eta_q(\lambda)$. An effective aperture area of the detection telescope is A_D and an overall efficiency of the detection system is represented by η_D. The solar flux spectral intensity at the satellite (before reflection) is denoted by $\Phi_S(\lambda)$, while the reflection efficiency and gain are η_R and G_R respectively. The wavelength-dependent atmospheric transmittance over the satellite - telescope slant range is expressed by $\eta_A(\lambda)$. A detailed analysis of these factors and their impact on the observed photon count rate is conducted by Kucharski et al. [30].

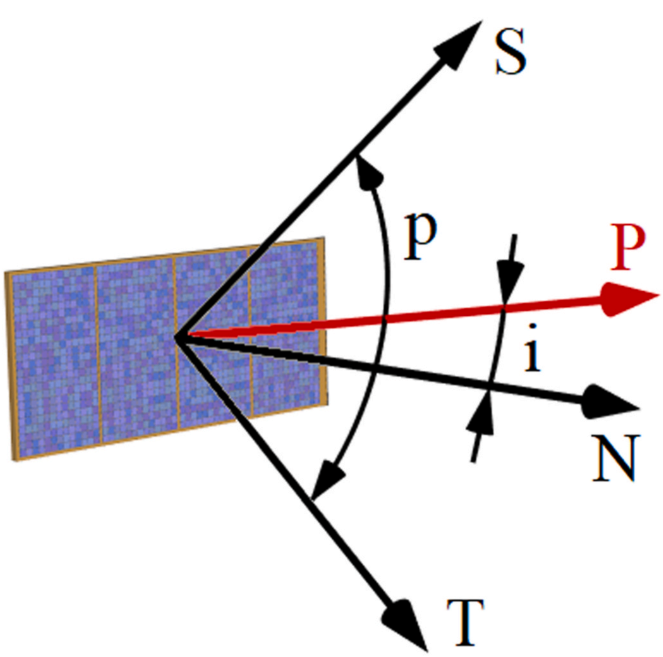

Fig. 2. Illustration of the reflection geometry with respect to the surface normal vector $\boldsymbol{N}$. Indicated are the unit direction vectors towards the Sun ($\boldsymbol{S}$) and the receiver telescope ($\boldsymbol{T}$), $\boldsymbol{P}$: phase vector, p: phase angle, i: inclination angle.

2.3. Photon flux mapping with Unit Phase Vector Field

Physical characteristics of the satellite-observer optical link is defined by the properties of the specular and diffuse fractions of solar reflection. Previous experiments [30 - section 3.3] prove that specularly reflective surface elements can deliver flux intensities of 2–3 orders of magnitude higher than the diffuse fraction of the link and thus strongly marking the occurrence of geometrical configuration when a phase vector $\boldsymbol{P}$ approaches or coincides with a surface normal vector $\boldsymbol{N}$ (Fig. 2). The inertial, equatorial coordinates (J2000) of a satellite-centered unit phase vector $\boldsymbol{P}$ can be calculated from a satellite position defined by a Two-Line Element set and the inertial coordinates of the Sun and an observer.

The light-curve analysis can be conveniently performed in a right-handed Cartesian, satellite-centered and -fixed Body Coordinate System (BCS) with the $+Z_{BCS}$ axis pointing towards the top of the structure as presented in Fig. 3. The transformation of a satellite-centered vector $\boldsymbol{r}$ from the Inertial Coordinate System, ICS (Earth-centered, J2000) to the satellite-fixed BCS can be defined as $\boldsymbol{r}_{BCS} = \boldsymbol{A}\,\boldsymbol{r}_{ICS}$, where the attitude tensor $\boldsymbol{A}$ between the inertial and the embedded rigid body frame is a rotation matrix computed by:

$$\boldsymbol{A} = \boldsymbol{R}_2(-x_P)\boldsymbol{R}_1(-y_P)\boldsymbol{R}_3(\gamma)\boldsymbol{R}_1\left(\frac{\pi}{2}-\delta\right)\boldsymbol{R}_3\left(\frac{\pi}{2}+\alpha\right) \tag{4}$$

The $\boldsymbol{R}_1$, $\boldsymbol{R}_2$ and $\boldsymbol{R}_3$ are the standard rotation matrices about the x, y and z-axis respectively, in a right-handed Cartesian coordinate system. The satellite inertial orientation is represented by right ascension α and declination δ of an angular momentum vector $\boldsymbol{L}$ about which the body spins in the direction indicated by the right-hand rule with an instantaneous rotational angle of γ. The pole coordinates x_P and y_P describe a relative position of $\boldsymbol{L}$ with respect to the satellite body axis $+Z_{BCS}$. The attitude tensor $\boldsymbol{A}$ is a continuous function of time-dependent parameters that allows converting the inertial direction vectors $\boldsymbol{S}$ and $\boldsymbol{T}$ to BCS and forming the Unit Phase Vector Field (UPVF) – a hypothetical construct in the satellite-fixed reference frame.

The body-fixed orientation of satellite surface elements (and their

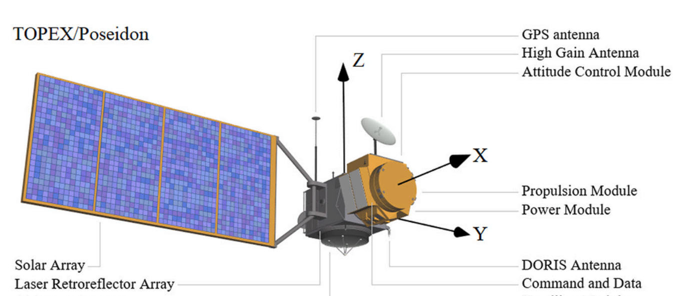

Fig. 3. TOPEX/Poseidon structural model and the satellite body-centered and -fixed Cartesian coordinate system.

normal vectors) defines the direction of the high- and low-intensity reflections annotated by the signal and noise markers (Fig. 4). The high-intensity signal markers for a box-wing type satellite are located in the equatorial plane of the body at the azimuthal coordinates of 0°, 90°, 180° and 270°. It is expected that the normal vectors to the front and back surfaces of the T/P solar panel are aligned with X_{BCS} thus corresponding to the equatorial signal markers at 0° and 180° respectively. A normal vector to an external surface of the Attitude Control Module points along $+Z_{BCS}$, while the normals to the thermal louvers of the Power Subsystem and the Communications and Data Handling Module (C&DHM) of the satellite bus [33] lay within the ZY_{BCS} plane and are inclined at −30° to the body equatorial plane creating an additional pair of the signal markers at the azimuthal angles of ±90°. The Instrument Module of T/P hosts a retroreflector array (RRA) for SLR that is constructed as an angled ring (1.5 m diameter) fitted around the nadir-pointed altimeter antenna [34]. The RRA consists of 192 corner cube reflectors arranged in two rows with the entrance face normal vectors off-pointed from the array optical axis (former nadir direction) by 40°. This defines a ring-shaped signal marker at BCS elevation of −50°. The low-intensity noise markers (Fig. 4) are located at the body-fixed directions where no major surface elements are pointing to (edges of the structure). An acceptance radius of the conic markers is set to an angular size of solar disk as seen from 1 AU distance (0.53°).

2.4. Attitude determination with Quanta Photogrammetry

Quanta Photogrammetry (QPM) projects properties of the satellite-observer optical link, measured at the single-photon sensitivity level, onto the Unit Phase Vector Field (UPVF) through the inertial-to-body attitude transformation (Eq. (4)). Once the body-fixed UPVF model is constructed, radiant intensity of the observed photon flux can be projected onto the corresponding coordinates of a phase vector in BCS. In practice, to optimize the computational process, the recorded photon count events are binned into 10 ms time intervals and expressed as photon count rate in particles/second as a measure of the flux intensity; the 10 ms duration of the binning interval corresponds to ~0.3° of T/P spin motion. The flux intensity samples are time stamped with the mean epoch of a binning interval and projected onto UPVF creating a photogram of radiant properties defined by the spacecraft technical and optical characteristics. The photometric samples are corrected for the major, elevation dependent effects (satellite slant range, atmospheric attenuation) as well as the detection system efficiency according to Eq. (3) (this process is explained in detail in Ref. [30]). Unmodeled changes in the sky transparency due to variation of the aerosol, dust and water vapor concentration can randomly affect the photometric observations.

Fig. 5-A presents a photogram of T/P observed during a single overhead pass on July 9, 2015 (9 min duration, >100E6 single-photon detection events recorded) with attitude parameters predicted by the spin model published in Ref. [1]. The photogram is plotted with Mollweide and polar projections where the sigma normalized flux intensities refer to the mean level (standard scale). Visible displacement of the photometric samples, a characteristic twist of the pattern, indicates a discrepancy between the true (inertial) and the modelled attitude parameters (Eq. (4)).

Satellite attitude determination process maximizes the radiant Signal-to-Noise ratio (SNR) of a photogram, where the average signal and noise values are computed from photometric samples falling into the corresponding conic markers (as defined on Fig. 4). The radiant SNR is a highly non-linear function of satellite attitude parameters, with a global maximum located in the vicinity of the initially predicted attitude state. In order to locate the solution, a hyperparameter grid search across the subset of attitude parameters is performed at a constant step of 0.1° for α, δ, γ, 0.01° for x_p, y_p, and 0.001 s for spin period. For every combination of the parameters the UPVF model is updated, new photogram obtained, and the corresponding SNR evaluated.

Fig. 6 presents an example cross section through the hyperparameter search space for the investigated pass (Fig. 5) where the SNR changes significantly during the validation process. The combination of attitude parameters resulting in the highest SNR constitutes the solution and is used to generate the final photogram presented on Fig. 5-B. Fig. 5-C shows the same dataset plotted in a logarithmic color scale, where the reflection intensities above the mean level are colored red; such representation visually magnifies fainter specular components of the pattern.

A set of 8 passes was selected for the QPM-based attitude determination of T/P. Each pass consists of a continuous (uninterrupted by atmospheric conditions), good quality photometric time series that cover the BCS elevation range of the signal and noise conic markers. The obtained attitude parameters and their corresponding standard errors are presented in Table 1. The standard errors of the inertial parameters (α, δ, γ, T) are obtained with the nonlinear least-squares where the partial derivatives of the individual SNR values are calculated for each signal marker separately using a finite-difference method. The mean values of the standard errors are: $\overline{\sigma}_\alpha = 1.36°$, $\overline{\sigma}_\delta = 0.69°$, $\overline{\sigma}_\gamma = 1.71°$, $\overline{\sigma}_T = 6.49\,ms$. The mean position of the spacecraft pole (figure axis) with respect to the angular momentum vector is $x_p = -2.92°$, $RMS = 0.39°$ and $y_p = -1.96°$, $RMS = 0.33°$, thus an average pole offset is 3.52°.

The body-fixed tilt of the Solar Array Normal Vector (SANV, perpendicular to the front surface of the panel) is determined from a 2D Gaussian fit to the centrally located (Fig. 5-B), highly specular radiant blob that corresponds to a signal marker at the BCS coordinates of

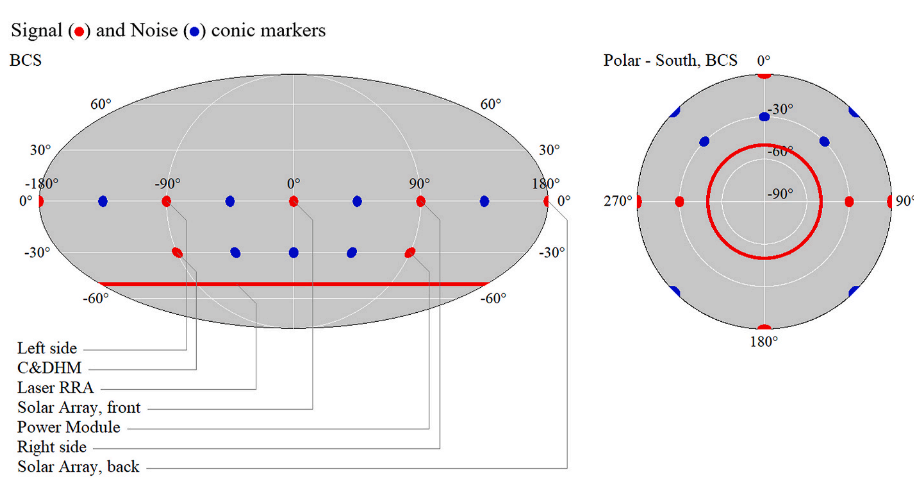

Fig. 4. Location of the radiant signal and noise conic markers in the satellite body-fixed coordinate system (BCS) in a Mollweide and (south) polar representations respectively.

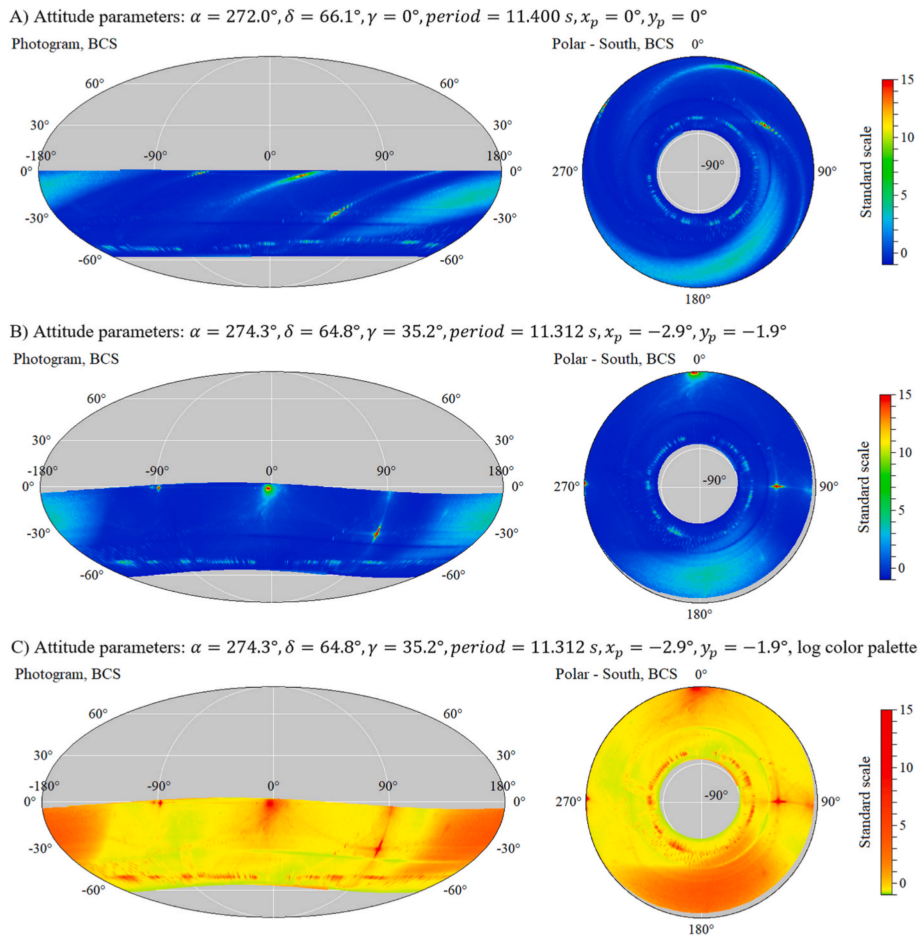

Fig. 5. The radiant photograms generated with a specific set of satellite attitude parameters: A) initial attitude prediction, B) resulting attitude parameters, C) same as B, but plotted with a logarithmic color scale. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

azimuth = 0°, elevation = 0° (Fig. 4). Locating the Gaussian peak in BCS gives the pitch-tilt angle of the solar panel elevation angle of SANV at −1.7°, RMS = 0.63° (8-pass mean).

2.5. Discussion on the QPM-based approach

The QPM finds the full set of satellite attitude parameters by optimizing the radiant SNR of a photogram. The hyperparameter grid search utilized in the process suffers from the curse of dimensionality, but is also perfectly parallel and can be efficiently executed by the GPU-powered supercomputing infrastructure of the Texas Advanced Computing Center (TACC) at the University of Texas at Austin [35].

Interpretation of the photograms is based on the rigid body assumption with fixed configuration of satellite surface elements; this is certainly true in the case of T/P and we can confirm that the radiant pattern geometry remains stable across multiple passes. The logarithmic plot (Fig. 5-C) enhances fine specular reflections that form distinctive features - more clearly visible on the polar plot: an elliptical curve with a long axis through the azimuth 0°/180° can be created by reflections off of the altimeter antenna [36], and the sharp edge that cuts off the high intensity reflection above azimuth 180° can be an evidence of the self-shadowing effect. An inverse analysis of a satellite radiant signature could lead to the development of more detailed shape estimation methods. While this paper is based on the measurements collected by a single telescope, the multiple, geographically distributed tracking systems would be needed to expand range of the satellite view directions and possibly eliminate the blind spots from the photograms.

The obtained attitude parameters agree with the SLR based results [1] and provide an additional degree of detail (the pole coordinates) that is used for better modelling of the physical forces acting on the body (Section 3). The determined coordinates of SANV confirm the initial assumption on the solar panel orientation in BCS.

Millidegree-scale errors in the inertial orientation of a unit phase vector are expected to arise from the limitations in the accuracy of the satellite position prediction with TLE, but do not have a detectable impact on the analysis accuracy levels presented in Table 1.

3. TOPEX/Poseidon spin-up due to direct solar photon pressure

3.1. Satellite rotational inertia

The physical mechanism of a dynamic interaction between T/P and the environmental forces and torques has been previously investigated [1,37] under simplified assumptions regarding the spacecraft configuration and the alignment of the spin and the body's pole axes. The QPM-based analysis delivers the body-fixed orientation of the solar array front surface (SANV) as well as the satellite pole coordinates (x_p, y_p) with respect to the angular momentum vector $\boldsymbol{L}$. The new findings are implemented as constants in the spin simulation algorithm with the satellite moment of inertia I being the variable solved-for during the best-fit process.

Dedicated SLR tracking campaign [1] has delivered accurate spin axis and spin rate observations that are used as the reference in the spin trend fitting process. A publicly available T/P macromodel provides the

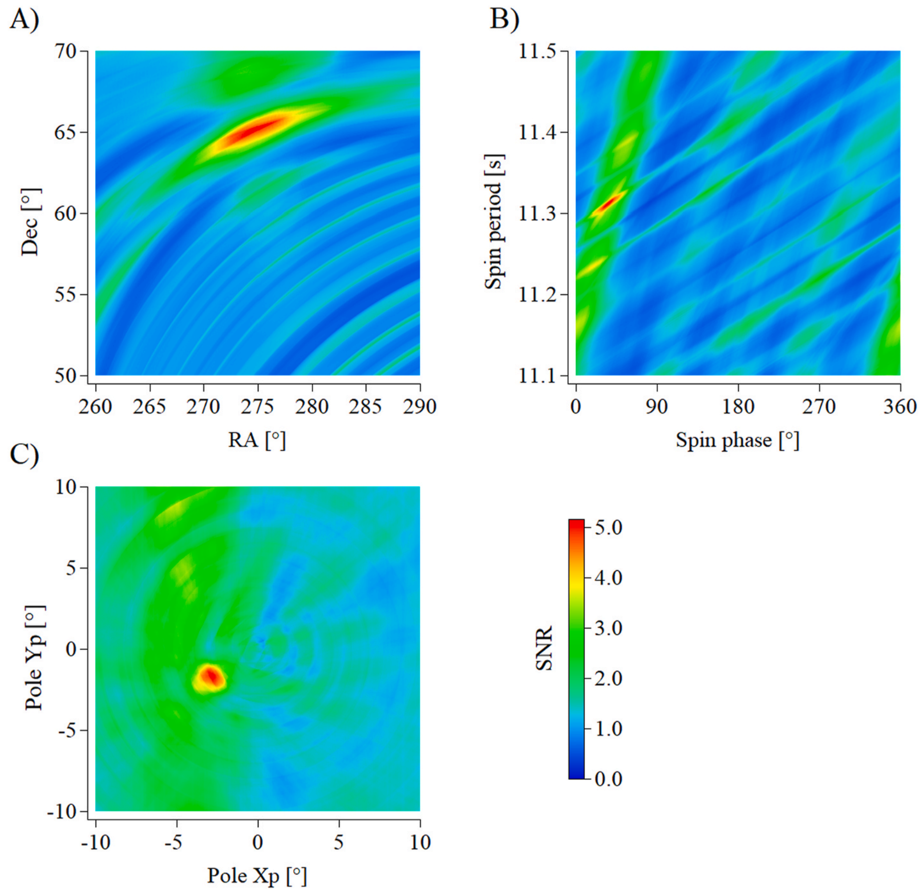

Fig. 6. Radiant SNR evaluation for the example pass (Fig. 5). A, B, C present cross-section through the search parameter space near the solution coordinates of $\alpha = 274.3^\circ$, $\delta = 64.8^\circ$, $\gamma = 35.2^\circ$, $T = 11.312\,s$, $x_p = -2.90^\circ$, $y_p = -1.93^\circ$.

technical dimensions and the optical coefficients that are utilized in simulation of the photon momentum transfer due to direct solar radiation, Earth-reflected sunlight (albedo) and Earth infrared emission. The elementary force vector $d\overline{F}$ caused on a surface element dS of a spacecraft by an incident photon flux Φ is given by Ref. [29]:

$$d\overline{\boldsymbol{F}} = -\frac{\Phi}{c}\left[(1-\rho)\widehat{\boldsymbol{R}} + 2\left(\frac{\delta}{3}+\rho\cos\beta\right)\widehat{\boldsymbol{n}}\right]dS|\cos\beta| \tag{5}$$

where c is the speed of light, ρ and δ represent the fraction of the incident light which is reflected and diffused respectively; the surface normal vector $\widehat{\boldsymbol{n}}$ creates angle β with the direction vector towards the radiation source $\widehat{\boldsymbol{R}}$. Due to the lack of information on the mass distribution and the magnetic properties of the body, the gravity-gradient and the eddy-currents precession torques are neglected here, but the actual inertial drift of the spin axis is taken into account and computed with the SLR-derived empirical model [1]. The spin simulations are performed with the 4th-order Runge-Kutta method where the photon pressure forces and torques acting on the elementary surface elements of the satellite macromodel are calculated and integrated.

The best fit between the predicted and observed spin period trends (Fig. 7-A) occurs for the satellite moment of inertia of $I = 69728$, $\sigma_I = 75.2$ $[kgm^2]$ with an RMS of the spin period residuals (observed minus calculated) of 19.2 ms. Changes in T/P spin period depend on the magnitude of the direct solar photon pressure torque in direction of the body's angular momentum vector $\boldsymbol{L}$ (Fig. 7-B, C). The simulations indicate correlation of the photon pressure torque and the spin period gradient with an angle between two vectors of the satellite-Sun direction and $\boldsymbol{L}$ (Fig. 7-D, E); on average, a one-degree increase of the Sun-L angle slows the spin-up rate by 56.4 μs/day. The simulated rotational state of T/P at the Sun-L angle near 116° gives a negligible component of the photon pressure torque projected on $\boldsymbol{L}$ which results in a stable spin motion.

3.2. Discussion on the photon interaction

The small in magnitude, but continuous, solar photon momentum transfer can explain the observed spin-up of T/P. The derived moment of inertia agrees with the theoretically possible range of values determined by the Debris SPin/Orbit Simulation Environment (D-SPOSE) [37].

The physical interaction between T/P and the environment is simulated with the optical coefficients of surface elements as defined at the beginning of the mission. However, the extreme ultra-violet (EUV) radiation and the atomic oxygen reactions at low Earth orbit enhanced by thermal cycling cause an inevitable degradation of the surface materials. The more complex observational definition of the satellite tumbling motion provided by QPM can better constrain the physical simulation scenario and open the possibility of researching the material aging effects through active modeling of the surface optical properties.

4. Conclusions and future work

Demonstrated for the first time, the QPM-based attitude determination of T/P delivers a more accurate description of its tumbling dynamics. It is now possible to measure the pole coordinates of the body with respect to the angular momentum vector as well as the body-fixed orientation of the highly specular front surface of the solar panel. This information benefits the physics-based simulations and allows deriving the satellite moment of inertia from the observed spin period time series. A more realistic attitude model can improve estimation of the growing

Table 1
The summary of attitude determination results. The inertial orientation of the angular momentum vector is given by α, δ; the inertial spin phase and spin period: γ, T. The body pole position with respect to the angular momentum vector is denoted by x_p, y_p.

Time	Angular momentum vector orientation (spin axis, inertial J2000)		Spin parameters (inertial)		Pole position	
UTC	α, σ_α [°]	δ, σ_δ [°]	γ, σ_γ [°]	T, σ_T [s, ms]	x_p [°]	y_p [°]
7-Jun-2015 00:14:00	295.3, 1.24	65.1, 0.68	287.2, 1.72	11.375, 6.1	−2.43	−2.50
24-Jun-2015 21:18:00	280.2, 1.30	63.2, 0.65	342.1, 1.68	11.344, 6.5	−3.41	−1.54
9-Jul-2015 23:16:00	274.3, 1.43	64.8, 0.72	35.2, 1.80	11.312, 7.3	−2.90	−1.93
10-Jul-2015 23:38:00	285.9, 1.52	65.6, 0.76	47.3, 1.74	11.306, 7.1	−2.44	−2.26
18-Mar-2016 20:12:00	26.2, 1.22	71.8, 0.69	278.4, 1.64	11.128, 6.8	−2.82	−1.94
24-Mar-2016 18:33:00	28.9, 1.46	69.2, 0.71	190.2, 1.76	11.120, 6.5	−3.28	−1.76
22-Oct-2018 16:44:00	229.3, 1.42	70.3, 0.67	158.0, 1.70	10.223, 5.9	−2.76	−2.14
25-Oct-2018 19:50:00	89.7, 1.30	70.2, 0.62	310.1, 1.63	10.205, 5.7	−3.35	−1.60

The mean pole position: $\bar{x}_p = -2.92°$, $RMS = 0.39°$ and $\bar{y}_p = -1.96°$, $RMS = 0.33°$.

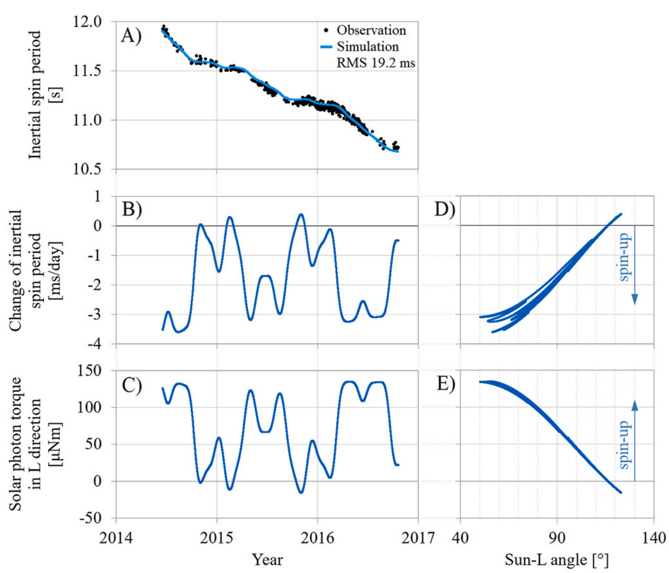

Fig. 7. The characteristics of T/P spin motion: A) observed (SLR, June 2014–October 2016) and simulated data, B) daily change of simulated spin trend, C) component of direct solar photon pressure torque in the direction of the satellite angular momentum vector *L*; D-E) the simulated results plotted versus an angle between *L* and the satellite-centered direction vector towards the Sun. Data plotted for sunlit condition only.

magnitude of centrifugal forces acting on T/P and prediction of the structural fragmentation risk of ageing spacecraft.

The potential of QPM will be further explored in an effort to carry out satellite shape recovery based on radiant photogram inversion with the existing models of the free-space optical link. Moreover, the possibility of using ultra-high repetition rate (MHz) Satellite Laser Ranging technology [2] will be investigated to sample retro-reflective properties of space objects, which can lead to the development of new type of quanta photograms for eclipsed satellites.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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